

Total No. of Questions : 6]

SEAT No. :

P123

APR. -16/BE/Insem. - 20

[Total No. of Pages : 3

B.E. (Mechanical Engineering)

INDUSTRIAL ENGINEERING

(2012 Course) (Elective-III) (Semester - II) (402049C)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answers should be written in one answer book.*
- 2) Answer Q.1 or Q.2, Q.3 or Q.4 or Q.5 or Q.6.*
- 3) Neat diagrams must be drawn wherever necessary.*
- 4) Figures to the right side indicate full marks.*
- 5) Assume suitable data, if necessary.*

Q1) a) Explain the contribution of Taylor and Gilberth to Industrial Engineering. **[6]**

b) The following information regarding the output produced and inputs consumed for a particular time period for a company given below. **[4]**

- i) Output = 10,000 Rs.
- ii) Human Input = 3,000 Rs.
- iii) Material Handling = 200 Rs.
- iv) Capital input = 300 Rs.
- v) Energy input = 100 Rs.
- vi) Others misc. Input = 50 Rs.

The values are in terms of based year rupee value. Compute various productivity indices, Total productivity.

OR

Q2) a) Explain briefly the factor's affecting the Productivity. **[6]**

b) There are two industries manufacturing two types of plugs. The standard time per piece is 1.5 minute output of the two industries is 300 & 200 respectively per shift of 8 hrs. then, **[4]**

- i) What is the Productivity of each per shift of 8 hrs?
- ii) What is the production of each per week (6 days) on the basis of double shift?

P.T.O.

- Q3) a)** What are therbligs give any five therbligs with symbols? [6]
b) Define work study. What are the components of work study? [4]

OR

- Q4) a)** Explain PMTS & MTM. [6]
b) Define Method study. Explain the Scope. [4]

- Q5) a)** Write short notes on (Any Three): [6]

- i) String Diagram.
- ii) Micro motion analysis.
- iii) SIMO chart.
- iv) Flow process chart.

- b) Following data relates to a work sampling study of a long cycle non-repetitive operation. [4]

- i) Total time of study = 10 days.
- ii) Total no. of observations = 1200.
- iii) Observations of Production activity = 900.
- iv) Manually controlled elements = 300.
- v) M/c Controlled elements = 300.
- vi) Total units Produce = 4000 pieces.
- vii) Rating index = 110%.

Observations of unavoidable delays = 100.

Calculate the standard time. Assuming that the study was made for 5 hours per day & the industry is running for single of 8 hrs. per day.

OR

- Q6)** a) What is Allowance? Explain any three types of allowances. [6]
- b) A work sampling study was conducted for 100 hrs. in the machine shop in order to estimate the standard time. The total number of observations recorded was 2500. No working activity could be noticed for 400 observations. The ratio between manual and machine elements was 2:1. Average rating factor was estimated as 1.15 and the total numbers of articles produced during the study period were 6000. Rest & personal allowance may be taken as 12% of the normal time. [4]

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